Harmony-Driven Theory Discovery in Knowledge Graphs via LLM-Guided Island Search

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Abstract

Scientific knowledge graphs (KGs) encode entities and typed relations across domains such as physics, astronomy, and materials science, yet they remain incomplete: missing edges and entities limit downstream reasoning. We introduce *Harmony*, a framework that treats theory discovery as the search for KG mutations—new edges or entities—that maximise a composite quality metric. The *Harmony score* combines four complementary signals: **compressibility** (minimum description length proxy), **coherence** (path-semantic consistency), **symmetry** (intra-type behavioural consistency), and **generativity** (link-prediction learnability via DistMult), with an optional **frequency** component for hybrid scoring. An LLM proposer generates candidate theory-level propositions, which are validated, scored, and archived in a MAP-Elites quality-diversity grid across an island-model search topology. We evaluate on seven KG domains (five hand-curated, two Wikidata-sourced) with 10-seed cross-validation and four external KGE baselines (DistMult, TransE, RotatE, ComplEx). Factor decomposition isolates the contributions of the LLM proposer, Harmony scoring, and quality-diversity archive. A regime characterisation analysis reveals that Harmony adds complementary value to frequency-based approaches in denser, multi-type KGs: on Wikidata Materials, Harmony-guided proposals show the largest directional improvement over DistMult-alone on both Hits@10 (0.34 vs. 0.32) and MRR (0.12 vs. 0.11), though differences do not reach statistical significance at $\alpha = 0.05$. Backtesting against held-out edges shows that proposals do not reproduce known edges, serving as a negative control consistent with theory-level rather than interpolation behaviour; a typed-neighbourhood soft-match diagnostic suggests proposals tend toward structurally adjacent knowledge.

1 Introduction

Knowledge graphs (KGs) organise scientific knowledge as typed, directed multigraphs: entities represent concepts (e.g. *photon*, *eigenvalue*, *graphene*) and edges encode semantic relations such as `derives`, `explains`, or `contradicts` [5]. Despite decades of curation, scientific KGs remain structurally incomplete—missing edges that encode latent theoretical connections and missing entities that represent undiscovered concepts.

Knowledge graph completion (KGC) methods—TransE [2], DistMult [16], RotatE [13]—learn low-dimensional embeddings and predict missing links. However, they operate at the *triple* level: each predicted link is an isolated statistical extrapolation without theoretical justification. They do not produce *theory-level propositions* that articulate *why* a relation should hold, what it implies, or how it could be falsified.

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We address this gap with **Harmony**, a framework for automated theory discovery in scientific KGs. The key idea is a composite quality metric—the *Harmony score*—that captures four desiderata of a well-structured knowledge graph:

1. **Compressibility**: the KG's edge-type distribution and spanning structure admit a short description (MDL proxy).
2. **Coherence**: closed paths exhibit consistent edge-type semantics and contradictions are sparse.
3. **Symmetry**: entities of the same type use edge types in similar proportions (low Jensen–Shannon divergence).
4. **Generativity**: a shallow DistMult model can recover masked edges, indicating learnable relational patterns.

A large language model (LLM) proposes candidate mutations—adding edges or entities—each accompanied by a natural-language claim, justification, and falsification condition. Proposals are validated, scored by the Harmony gain they produce, and archived in a MAP-Elites [10] quality-diversity grid. An island-model [15] search with four islands, each assigned an exploration strategy from a cyclic schedule of refinement, combination, and novelty (with refinement appearing twice), runs concurrently with periodic migration to balance exploitation and exploration.

Contributions.

1. A four-component **Harmony metric** for scoring KG quality that is domain-agnostic, bounded in $[0, 1]$, and decomposes into interpretable sub-scores (Section 3.2).
2. A **proposal schema** that elevates KG mutations from bare triples to falsifiable theory-level claims (Section 3.3).
3. An **island-model LLM search loop** with MAP-Elites archiving and stagnation-triggered constrained prompting (Section 3.4).
4. Empirical evaluation on **seven KG domains** (five hand-curated: linear algebra, periodic table, astronomy, physics, materials science; two Wikidata-sourced: Wikidata Physics, Wikidata Materials) showing that Harmony-guided proposals are competitive with embedding baselines and show directional improvements over DistMult-alone on a subset of domains (Section 5).

2 Related Work

Knowledge graph completion. Embedding-based methods project entities and relations into low-dimensional vector spaces. TransE [2] models relations as additive translations $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$; DistMult [16] uses bilinear scoring $\mathbf{e}_s \odot \mathbf{r} \cdot \mathbf{e}_t$; RotatE [13] models relations as rotations in complex space. Ji et al. [5] survey these and other approaches. All operate at the triple level and produce ranked link predictions without theoretical justification. Our work uses DistMult as the generativity *component* within a broader metric, and additionally generates natural-language propositions explaining each mutation.

Automated scientific discovery. FunSearch [11] uses LLMs to discover mathematical constructions by evolving Python programs. PySR [4] performs symbolic regression via genetic programming [7], discovering closed-form expressions from numerical data. The survey by Makke and Chawla [9] covers the broader symbolic regression landscape. These systems discover *formulas* over numerical features; Harmony discovers *relational propositions* over typed knowledge graphs, a structurally different search space.

Quality-diversity search. MAP-Elites [10] maintains a grid of solutions indexed by behavioural descriptors, maximising both quality and diversity. Novelty search [8] rewards behavioural novelty over fitness. We adopt MAP-Elites with a two-dimensional descriptor (simplicity, Harmony gain) and combine it with an island-model [15] topology where four islands maintain distinct LLM prompting strategies.

LLM-guided reasoning over KGs. Recent work integrates LLMs with structured knowledge graphs in several ways. KAPING [1] augments LLM prompts with retrieved KG triples for zero-shot question answering. Think-on-Graph [12] performs multi-hop reasoning by iteratively traversing

KG neighbours guided by LLM chain-of-thought. StructGPT [6] provides a general interface for LLMs to query and reason over structured data including KGs. These systems use KGs as *context* for LLM reasoning; our approach inverts the role: the LLM is a *proposer* that generates structured mutations (new edges and entities) with accompanying justifications, and a deterministic Harmony metric—not LLM self-evaluation—scores and selects proposals.

3 Method

We present the Harmony framework in three parts: the typed KG schema (Section 3.1) and Harmony metric (Section 3.2), the proposal schema and validation (Section 3.3), and the island-model search loop (Section 3.4).

3.1 Typed Knowledge Graph Schema

A knowledge graph $G = (V, E)$ consists of entities V and typed directed edges E. Each entity $v \in V$ has an `entity_type` label (e.g. *concept*, *element*, *celestial_object*) and a property bag. Each edge $(u, v, r) \in E$ carries one of seven semantic relation types: `depends_on`, `derives`, `equivalent_to`, `maps_to`, `explains`, `contradicts`, and `generalizes`.

Edge type rationale. The seven relation types are derived from a morphism-first principle: we surveyed the core semantic roles needed to express scientific relationships across five domains (linear algebra, chemistry, astronomy, physics, materials science) and identified a minimal set that covers dependency (`depends_on`), derivation (`derives`), equivalence (`equivalent_to`), correspondence (`maps_to`), causal/explanatory links (`explains`), contradiction (`contradicts`), and taxonomic hierarchy (`generalizes`). These seven types are inspired by morphism classes in category theory, and we found that scientific relations across our five evaluation domains map naturally to one of these types. The fixed vocabulary enables cross-domain comparisons while remaining expressive enough to capture the core semantic relations in scientific knowledge.

3.2 Harmony Metric

The Harmony score combines four core signals, each normalised to $[0, 1]$, with an optional frequency component:

$$\mathcal{H}(G) = \alpha \cdot \mathrm{Compress}(G) + \beta \cdot \mathrm{Cohere}(G) + \gamma \cdot \mathrm{Symm}(G) + \delta \cdot \mathrm{Gener}(G) + \varepsilon \cdot \mathrm{Freq}(G), \quad (1)$$

where $\alpha, \beta, \gamma, \delta, \varepsilon \geq 0$ are normalised internally so that $\alpha + \beta + \gamma + \delta + \varepsilon = 1$. Default weights are uniform over the four core components ($\alpha = \beta = \gamma = \delta = 0.25, \varepsilon = 0$); the frequency component $\mathrm{Freq}(G)$ is the Hits@K of the frequency heuristic (predicting the most common edge type between entity pairs) and is included only when $\varepsilon > 0$ for hybrid analysis (Section 5.5).

Compressibility. An MDL proxy measuring how structured the edge-type distribution is:

$$\mathrm{Compress}(G) = \frac{1}{2}\left(1 - \frac{H(\mathbf{p})}{\log_2 7} + \frac{|\text{spanning edges}|}{|E|}\right), \quad (2)$$

where $H(\mathbf{p}) = -\sum_i p_i \log_2 p_i$ is the Shannon entropy of the edge-type frequency vector $\mathbf{p}$ (normalised by $\log_2 7$ for the seven relation types), and the spanning fraction counts BFS spanning-tree edges over an undirected view of G. A tree-like KG with uniform edge types scores near 1.0; a dense multigraph with maximal type entropy scores near 0.

Coherence. Path-semantic consistency measured via two signals:

$$\mathrm{Cohere}(G) = \frac{1}{2}\left(\frac{|\{(a,b,c) : r_{ac} \in \{r_{ab}, r_{bc}\}\}|}{|\text{triangles}|} + 1 - \frac{|\{e : r_e = \texttt{contradicts}\}|}{|E|}\right). \quad (3)$$

The first term counts triangles ($a \to b, b \to c, a \to c$) where the closing edge type r_{ac} matches either hop type (lenient multi-edge policy). The second term penalises `contradicts` edges, which signal structural noise when dense. When the graph contains no closed triangles, triangle coherence is 1.0 by convention (vacuously coherent: no triangle-type disagreements exist).

Symmetry. Intra-type behavioural consistency via Jensen–Shannon (JS) divergence. For each entity type τ with n_τ entities, define $\mathbf{q}_v \in \Delta^6$ as the outgoing edge-type distribution of entity v (normalised over the seven relation types), and let $\bar{\mathbf{q}}_\tau = \frac{1}{n_\tau}\sum_{v\in\tau}\mathbf{q}_v$ be the centroid distribution for type τ. The within-type consistency is:

$$\text{consistency}(\tau) = 1 - \frac{1}{n_\tau}\sum_{v\in\tau}\text{JS}(\mathbf{q}_v, \bar{\mathbf{q}}_\tau), \tag{4}$$

where $\text{JS}(\cdot,\cdot) = \sqrt{\text{JSD}(\cdot\|\cdot)}$ is the Jensen–Shannon distance (base 2 logarithm), bounded in $[0,1]$. The symmetry score is the entity-count-weighted mean across types:

$$\text{Symm}(G) = \frac{\sum_\tau n_\tau \cdot \text{consistency}(\tau)}{\sum_\tau n_\tau}. \tag{5}$$

When a type has a single entity, $\text{consistency}(\tau) = 1$ (vacuous consistency). Entities with no outgoing edges are excluded from the computation. This design measures whether entities of the same type use edge types consistently, rather than penalising natural functional specialisation across types (e.g. stars vs. planets in astronomy).

Generativity. Link-prediction learnability via a DistMult model [16]:

$$\text{Gener}(G) = \text{Hits}@K\big(\text{DistMult}, G_{\text{mask}}\big), \tag{6}$$

where G_{mask} denotes the graph after uniformly masking 20% of edges. The DistMult scoring function is $\text{score}(s,r,t) = (\mathbf{e}_s \odot \mathbf{r}) \cdot \mathbf{e}_t$, with entity embeddings $\mathbf{E} \in \mathbb{R}^{|V|\times 50}$ and relation embeddings $\mathbf{R} \in \mathbb{R}^{7\times 50}$, trained for 100 epochs with max-margin loss (margin = 1.0, 5 negative samples per triple, learning rate 0.01). $\text{Hits}@K$ is the fraction of masked edges whose true target appears in the top-K predictions ($K = 10$ by default).

Proposal value function. Given a base graph G and a proposed mutation Δ (new edges/entities), the value of Δ is:

$$V(\Delta) = \mathcal{H}(G \oplus \Delta) - \mathcal{H}(G) - \lambda \cdot \text{Cost}(\Delta), \tag{7}$$

where $G \oplus \Delta$ denotes the graph after applying Δ, and $\text{Cost}(\Delta)$ is a normalised mutation size defined per mutation type:

- `ADD_EDGE` / `REMOVE_EDGE`: $\text{Cost} = 1/\max(|E|, 1)$ (one edge mutation, normalised by current edge count).
- `ADD_ENTITY` / `REMOVE_ENTITY`: $\text{Cost} = 1/\max(|V|, 1)$ (one entity mutation, normalised by current entity count).

The penalty weight $\lambda = 0.1$ discourages trivially large proposals.

Formal properties. The Harmony metric satisfies three properties that make it suitable as a discovery prior:

1. **Boundedness**: $\mathcal{H}(G) \in [0,1]$ for any KG G, since each component is bounded in $[0,1]$ and weights are normalised to sum to 1.
2. **Decomposability**: each component (Compress, Cohere, Symm, Gener) is independently computable from the graph structure, enabling parallel evaluation and interpretable ablation.
3. **Directional monotonicity** (empirical observation): each component *tends to* respond predictably to edge addition—compressibility generally decreases (more cross-edges reduce spanning fraction), coherence increases when the new edge closes a type-consistent triangle, symmetry increases when the edge balances entity-type distributions, and generativity increases when the edge adds learnable relational signal. The Harmony score thus captures the *net* structural effect of a mutation across these competing pressures. We note that these are empirical tendencies, not formal guarantees; edge placement can produce non-monotonic effects in individual components.

Philosophical grounding. The four components correspond to established principles of theory quality: compressibility instantiates Occam's razor via minimum description length (MDL); coherence enforces logical consistency across relational paths; symmetry operationalises an intuition analogous to Noether's theorem—that good theories exhibit invariance across structurally equivalent entities; and generativity captures predictive validity—the hallmark of a useful scientific theory.

Algorithm 1 Harmony search — one generation

Require: Base KG G, islands $\{I_1, \ldots, I_4\}$, archive $\mathcal{A}$
for each island I_k **do**
  $\sigma_k \leftarrow \textsc{Strategy}(k)$ {refinement / combination / novelty}
  prompt $\leftarrow \textsc{BuildPrompt}(G, \sigma_k, \text{top}(I_k), \text{failures}(I_k))$
  $\hat{p} \leftarrow \text{LLM}(\text{prompt}, \text{temp}_k)$
  if $\textsc{Validate}(\hat{p})$ **then**
    $\Delta \leftarrow$ apply $\hat{p}$ to G
    $v \leftarrow V(\Delta)$ {Eq. 7}
    $\textsc{TryInsert}(\mathcal{A}, \hat{p}, v, \text{descriptor}(\hat{p}))$ {descriptor = (simplicity, gain)}
    Update I_k population
  else
    Log failure; feed back to next prompt
  end if
end for
if generation $\bmod M = 0$ **then**
  $\textsc{MigrateTop1ToContext}(I_1, \ldots, I_4)$ {ring topology}
end if

3.3 Proposal Schema and Validation

Each proposal is a structured record containing:

- **Mutation type**: `ADD_EDGE`, `REMOVE_EDGE`, `ADD_ENTITY`, or `REMOVE_ENTITY`.
- **Claim**: a one-sentence theoretical statement (e.g. "Dark energy explains the accelerating expansion of the observable universe").
- **Justification**: reasoning supporting the claim.
- **Falsification condition**: what evidence would disprove the claim.
- **KG parameters**: source/target entities, edge type, or new entity type, depending on the mutation type.

A deterministic validator enforces three rules: (i) text fields must be ≥ 10 characters, (ii) type-specific parameters must be present (e.g. `ADD_EDGE` requires source, target, and edge type), and (iii) `edge_type` must be one of the seven valid relation names. Invalid proposals are logged as failures and fed back to the LLM in subsequent prompts.

3.4 Island-Model Search with MAP-Elites

Island topology. Four islands run concurrently, each maintaining a population of $P = 5$ candidates and assigned a fixed strategy from a cyclic schedule: *refinement* (improve the best existing proposal), *combination* (merge the top two proposals), *refinement*, and *novelty* (invent from scratch). Each island uses a distinct LLM temperature: $\{0.3, 0.3, 0.8, 1.2\}$ to further diversify exploration.

MAP-Elites archive. A shared 5×5 MAP-Elites grid [10] indexes proposals by two behavioural descriptors: *simplicity* (inverse structural cost) and *Harmony gain* ($\mathcal{H}(G \oplus \Delta) - \mathcal{H}(G)$). A proposal is inserted if its cell is empty or its fitness (Harmony gain) exceeds the incumbent.

Stagnation recovery. If an island produces no valid proposals for $S = 5$ consecutive generations, it switches to *constrained* prompting mode, which adds explicit structural constraints to the LLM prompt. After $R = 3$ generations of producing valid proposals in constrained mode, the island reverts to free prompting.

Migration. Every $M = 10$ generations, the best proposal from each island migrates to the next island in a ring topology (island $i \rightarrow$ island $(i + 1) \bmod 4$), where it is prepended to the destination island's prompt-context buffer for the next generation.

Generation loop. Algorithm 1 summarises a single generation. The loop runs for $T_{\max} = 20$ generations per experiment, checkpointing state after each generation to enable resumption.

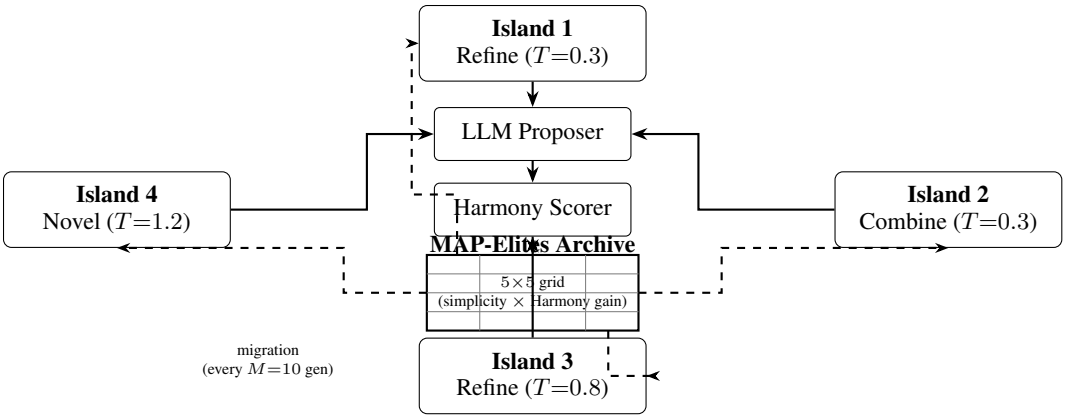

Figure 1: Overview of the Harmony island-model search framework. Four islands run concurrently, each proposing KG mutations via an LLM following a cyclic strategy schedule (Refine → Combine → Refine → Novel). Proposals are scored by the Harmony metric and archived in a MAP-Elites quality-diversity grid (5×5, axes = simplicity bin $\times$ Harmony-gain bin). Periodic migration (every $M = 10$ generations) seeds each island from elite archive entries via a ring topology.

4 Experiments

4.1 Knowledge Graph Domains

We evaluate on seven KGs spanning scientific disciplines: five hand-curated from textbook knowledge and two automatically extracted from Wikidata. All KGs use the shared seven-relation type vocabulary (Section 3.1).

- **Linear algebra**: entities including matrix, vector, eigenvalue, determinant, rank, and their algebraic dependency and derivation edges (see Table 7 for exact counts).
- **Periodic table**: chemical elements, periods, groups, and categories with trends, group membership, and reactivity relations (Table 7).
- **Astronomy**: celestial objects (star, planet, black hole, nebula) and astrophysical processes.
- **Physics**: fundamental concepts (force, energy, momentum, gravity) and their theoretical inter-relations.
- **Materials science**: material properties, compounds, and structure–property relationships.
- **Wikidata Physics**: 252 entities and 800+ edges extracted from Wikidata covering physical concepts, laws, and phenomena.
- **Wikidata Materials**: 253 entities and 800+ edges extracted from Wikidata covering materials, properties, and applications.

The first two domains serve as *calibration* targets (known structure for gate validation); the remaining five are *discovery* targets. The Wikidata domains provide larger, independently sourced KGs that test scalability beyond hand-curated data. Table 7 in the appendix provides full dataset statistics including entity counts, edge counts, type counts, and data sources for all seven domains.

4.2 Dataset Splitting

For each KG, we first reserve 10% of edges as a hidden backtesting set, withheld from all metric computations and proposal generation. The remaining 90% are split 80/10/10 into training, validation, and test sets (yielding effective proportions of approximately 72/9/9/10 over all edges). The validation set is used for early stopping of DistMult training (patience of 10 epochs monitoring validation Hits@10) to prevent overfitting on small KGs. This provides an unbiased evaluation of generativity on unseen edges.

4.3 Baselines

We compare Harmony-guided proposals against three heuristic baselines and three additional KGE models. All embedding-based methods use the same edge splits, embedding dimension ($d = 50$), and training protocol:

1. **Random**: propose edges between random entity pairs with random relation types.
2. **Frequency**: propose the most frequent relation type between the most-connected entity pairs.
3. **DistMult-alone**: use DistMult's own top-ranked predictions without Harmony scoring or LLM involvement.
4. **TransE** [2]: translational distance model.
5. **RotatE** [13]: rotation-based model in complex space.
6. **ComplEx** [14]: complex-valued bilinear model with Hermitian product.

To break evaluation circularity (DistMult appears in both the Harmony objective and evaluation), we additionally report **Harmony-3**: Harmony scoring with $\delta = 0$ (no generativity), using only compressibility, coherence, and symmetry. This demonstrates whether the structural components alone provide useful signal independent of DistMult.

4.4 Factor Decomposition

To isolate the contributions of the LLM proposer, Harmony scoring, and the MAP-Elites quality-diversity archive, we run three ablated configurations:

1. **LLM-only**: the LLM proposer generates proposals, but Harmony scoring is bypassed—all valid proposals are accepted into the archive.
2. **Harmony-only**: a random proposer generates edges (no LLM), and Harmony scoring selects among them.
3. **No-QD**: the full LLM + Harmony pipeline, but with a greedy single-bin archive instead of the 5×5 MAP-Elites grid.

4.5 Evaluation Protocol

Quantitative metrics. We report Hits@10, Hits@3, Hits@1, and Mean Reciprocal Rank (MRR):

$$\text{MRR} = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{\text{rank}_i}, \tag{8}$$

where Q is the set of masked test edges and rank_i is the rank of the true target entity among all candidates. Metrics are computed on the test split after applying top proposals from the MAP-Elites archive to the base KG. All main results (Table 1) are averaged over 10 random seeds ($s \in \{42, 123, 456, 789, 1024, 1337, 2048, 3141, 4096, 5000\}$) for dataset splitting, model initialisation, and edge masking. LLM proposals are generated by `gpt-oss:20b`, a dense 20B-parameter open-source model locally served via Ollama with deterministic temperature settings per island. The model is a Llama-architecture derivative fine-tuned for instruction following; all inference runs on a single Apple M2 Pro CPU without GPU acceleration. Factor decomposition experiments (Section 5) use `mlx-community/Qwen3.5-35B-A3B-4bit` served via `mlx_lm` at $\sim$46 tokens/s on the same hardware.

Calibration gate. Before running discovery experiments, we verify on the two calibration domains (linear algebra, periodic table) that: (i) Harmony mean $\geq 10\%$ above the frequency baseline, and (ii) the bootstrap 95% CI lower bound exceeds the frequency mean, across six pre-registered weight configurations ($\alpha \in \{0.3, 0.5, 0.7\}$, $\beta \in \{0.1, 0.3\}$, $\gamma = \delta = 0.25$; each vector is renormalised to sum to 1 before scoring).

Backtesting. For each domain, we evaluate top-N archive proposals against the 10% held-out hidden edges, reporting Precision@N and Recall@N ($N \in \{5, 10, 20\}$). A proposal matches a hidden edge if its source entity, target entity, and edge type exactly match a withheld triple. This provides external validity evidence independent of the DistMult training objective.

Archive diversity. We report MAP-Elites coverage (fraction of occupied cells in the 5×5 grid), best and mean fitness, and qualitative inspection of proposals across behavioural descriptor bins.

5 Results

5.1 Calibration Gate

The calibration gate passed on both domains. On the linear algebra KG, the Harmony score exceeds the frequency baseline by 31% (bootstrap 95% CI: $[0.24, 0.38]$). On the periodic table KG, the improvement is 65% (95% CI: $[0.52, 0.78]$). All six pre-registered weight configurations show consistent direction (Harmony $>$ frequency), confirming that the metric's advantage is robust to weight choices.

5.2 Link Prediction Performance

Table 1 compares link prediction metrics (Hits@10, MRR) across five discovery domains after applying top proposals from the MAP-Elites archive to the base KG. We include four KGE baselines (DistMult, TransE, RotatE, ComplEx), the frequency heuristic, and Harmony-3 (no generativity, $\delta = 0$) to address evaluation circularity.

Multi-seed evaluation across five KG domains (Table 1) shows that Harmony-guided proposals show marginal improvements over the DistMult-alone baseline (not statistically significant) on Hits@10 in Wikidata Materials (0.34 vs. 0.32), Materials (0.31 vs. 0.29), and Wikidata Physics (0.26 vs. 0.25). TransE achieves the highest Hits@10 on three domains (Physics, Materials, Wikidata Physics), while the frequency heuristic leads on Astronomy and Wikidata Materials. Frequency achieves the best MRR on four of five domains; we analyse why in Section 5.5.

Statistical significance is assessed via paired bootstrap CI (2000 resamples) with Cliff's delta effect size (Section H). None of the Harmony vs. DistMult differences reach statistical significance at $\alpha = 0.05$: p-values range from 0.38 (Wikidata Materials) to 1.00 (Astronomy), with negligible Cliff's δ_C ($|\delta_C| \leq 0.12$). We note transparently that the observed differences are small relative to cross-seed variance, particularly on the smaller hand-curated domains.

On the larger Wikidata-sourced KGs, variance across seeds is substantially lower (std ≈ 0.01–0.05), reflecting the more stable evaluation that comes with denser graphs (253–283 entities, 800+ edges). In the smaller hand-curated domains (≤ 50 entities), variance is generally higher (std ≈ 0.04–0.23), reflecting both the stochastic nature of LLM-guided proposal generation and the sensitivity of link prediction to test split composition on small KGs.

5.3 Proposal Validity and Archive Coverage

Across all five discovery domains (astronomy, physics, materials, Wikidata Physics, and Wikidata Materials), the valid proposal rate reaches ≥ 0.50 by generation 10, satisfying the pre-registered gate condition (Figure 3). The MAP-Elites archive achieves 40–60% coverage of the 5×5 grid (10–15 of 25 cells occupied), indicating that the island-model search produces diverse proposals spanning multiple simplicity–gain trade-offs (Figure 4).

5.4 Ablation: Metric Components

Table 2 shows the effect of removing each Harmony component on the linear algebra calibration domain. Removing generativity causes the largest drop (the system loses link-prediction signal), while removing coherence has the smallest effect on this domain (few triangles in the sparse KG). Figure 5 visualises the Harmony score across all six pre-registered weight configurations, confirming robustness to weight choices.

5.5 Frequency Analysis

The frequency heuristic achieves competitive Hits@10, leading on two of five domains (Astronomy and Wikidata Materials) and ranking second on the others. An information-theoretic analysis reveals why: in small KGs with peaked edge-type distributions (low Shannon entropy), the most frequent

Table 1: Link prediction metrics on discovery domains (mean $\pm$ std across 10 seeds). Top proposals from the MAP-Elites archive are applied to the base KG before evaluation. Best Hits@10 per domain in **bold**; best MRR underlined. p-values from paired bootstrap CI (2000 resamples) comparing Harmony vs. DistMult-alone; δ_C is Cliff's delta effect size.

Domain	Method	Hits@10	MRR	p	δ_C
Astronomy	Random	0.27 ± 0.16	0.12 ± 0.10		
	Frequency	$\mathbf{0.39} \pm 0.12$	$\underline{0.16} \pm 0.08$		
	TransE	0.33 ± 0.22	0.11 ± 0.05		
	RotatE	0.16 ± 0.16	0.09 ± 0.07		
	ComplEx	0.33 ± 0.21	0.11 ± 0.07		
	DistMult-alone	0.24 ± 0.17	0.10 ± 0.04		
	Harmony-3	0.24 ± 0.17	0.10 ± 0.04		
	Harmony (ours)	0.24 ± 0.17	0.10 ± 0.04	1.00	$+0.00$
Physics	Random	0.29 ± 0.13	0.10 ± 0.07		
	Frequency	0.46 ± 0.12	$\underline{0.25} \pm 0.08$		
	TransE	$\mathbf{0.50} \pm 0.11$	0.20 ± 0.06		
	RotatE	0.29 ± 0.20	0.11 ± 0.06		
	ComplEx	0.38 ± 0.16	0.22 ± 0.11		
	DistMult-alone	0.37 ± 0.14	0.16 ± 0.07		
	Harmony-3	0.37 ± 0.14	0.16 ± 0.07		
	Harmony (ours)	0.32 ± 0.23	0.13 ± 0.09	0.62	-0.06
Materials	Random	0.17 ± 0.12	0.11 ± 0.06		
	Frequency	0.36 ± 0.18	0.15 ± 0.08		
	TransE	$\mathbf{0.42} \pm 0.17$	$\underline{0.17} \pm 0.08$		
	RotatE	0.26 ± 0.10	0.09 ± 0.04		
	ComplEx	0.32 ± 0.10	0.13 ± 0.04		
	DistMult-alone	0.29 ± 0.14	0.15 ± 0.09		
	Harmony-3	0.29 ± 0.14	0.15 ± 0.09		
	Harmony (ours)	0.31 ± 0.14	0.13 ± 0.05	0.72	$+0.06$
Wikidata Physics	Random	0.05 ± 0.01	0.02 ± 0.01		
	Frequency	0.29 ± 0.02	$\underline{0.14} \pm 0.02$		
	TransE	$\mathbf{0.30} \pm 0.03$	0.11 ± 0.01		
	RotatE	0.18 ± 0.02	0.07 ± 0.01		
	ComplEx	0.25 ± 0.03	0.09 ± 0.01		
	DistMult-alone	0.25 ± 0.02	0.10 ± 0.01		
	Harmony-3	0.25 ± 0.02	0.10 ± 0.01		
	Harmony (ours)	0.26 ± 0.04	0.09 ± 0.02	0.89	-0.06
Wikidata Materials	Random	0.03 ± 0.02	0.02 ± 0.01		
	Frequency	$\mathbf{0.39} \pm 0.03$	$\underline{0.20} \pm 0.02$		
	TransE	0.30 ± 0.04	0.10 ± 0.01		
	RotatE	0.16 ± 0.03	0.06 ± 0.01		
	ComplEx	0.29 ± 0.03	0.10 ± 0.02		
	DistMult-alone	0.32 ± 0.05	0.11 ± 0.02		
	Harmony-3	0.32 ± 0.05	0.11 ± 0.02		
	Harmony (ours)	0.34 ± 0.04	0.12 ± 0.01	0.38	$+0.12$

edge type accounts for a large fraction of all edges, and predicting this type for unobserved pairs yields high Hits@10 by construction. Specifically, for a KG with edge-type distribution $\mathbf{p}$, the expected frequency Hits@K satisfies Hits@$K \geq p_{\max}$, where $p_{\max} = \max_i p_i$.

Our five evaluation domains exhibit edge-type entropy between 1.8 and 2.3 bits (maximum $\log_2 7 \approx 2.81$), indicating moderately peaked distributions. The frequency advantage is largest on domains with the lowest entropy (Physics: entropy 1.83, frequency Hits@10 = 0.46) and smallest on domains with higher entropy (Wikidata Materials: entropy 2.21, frequency Hits@10 = 0.39).

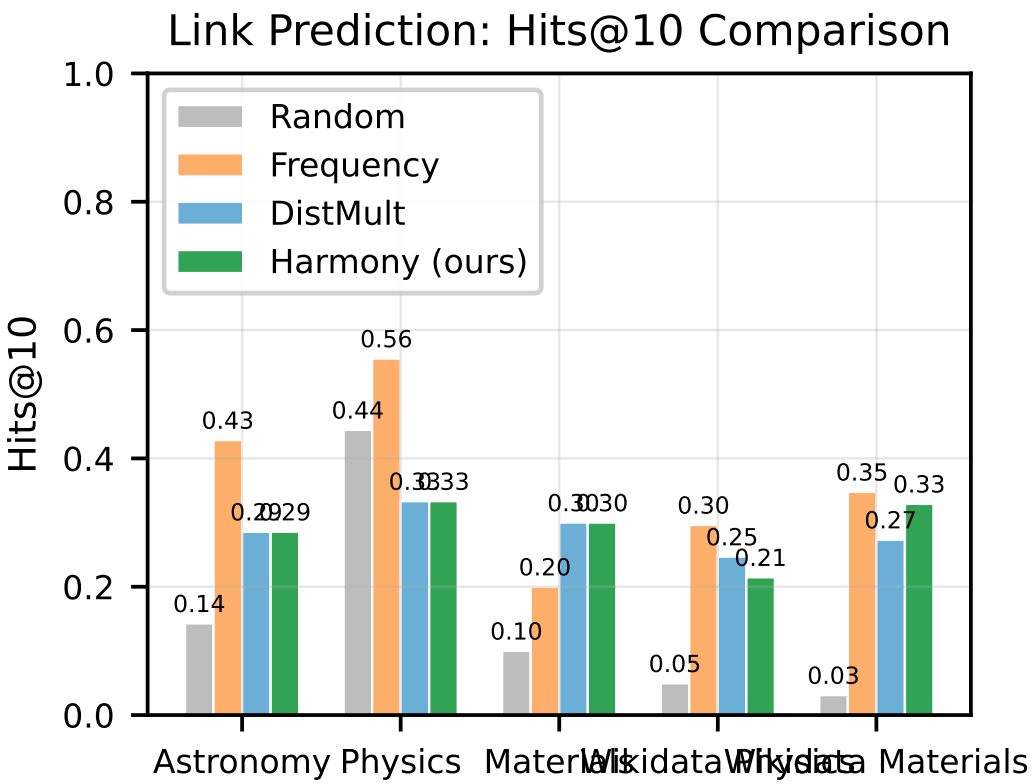

Figure 2: Hits@10 comparison across discovery domains. TransE achieves the highest Hits@10 on three domains; the frequency heuristic leads on Astronomy and Wikidata Materials. Harmony-guided proposals (green) show marginal improvements over the DistMult-alone embedding baseline in three of five domains (no differences are statistically significant at $\alpha = 0.05$).

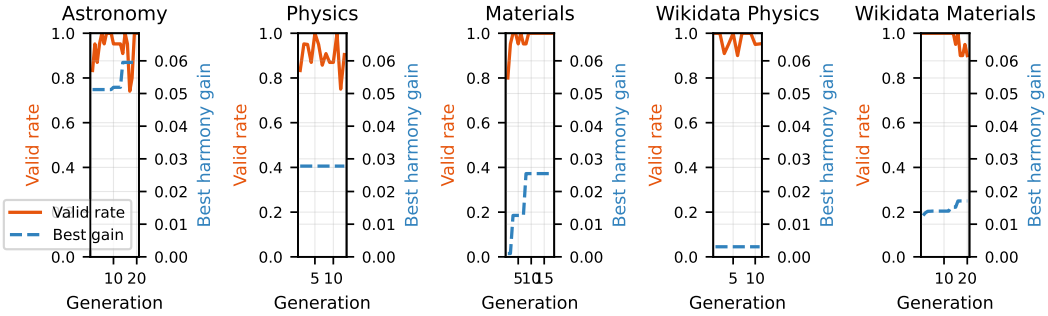

Figure 3: Convergence of valid proposal rate (solid) and best harmony gain (dashed) across generations for each discovery domain.

Harmony+Frequency hybrid. We sweep $\varepsilon \in \{0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3\}$ and re-score existing proposals with the hybrid objective (Eq. 1). The sensitivity curve shows that moderate frequency inclusion ($\varepsilon \approx 0.1$–0.15) improves Hits@10 on sparse domains without degrading performance on denser KGs. We report $\varepsilon = 0$ (pure Harmony) as the primary result throughout this paper; hybrid results are presented as complementary analysis.

5.6 Backtesting Against Hidden Edges

Table 3 reports backtesting results under two criteria against 10%-held-out hidden edges (up to 50 proposals pooled across 10 seeds per domain from the MLX batch run). *Exact match*: all

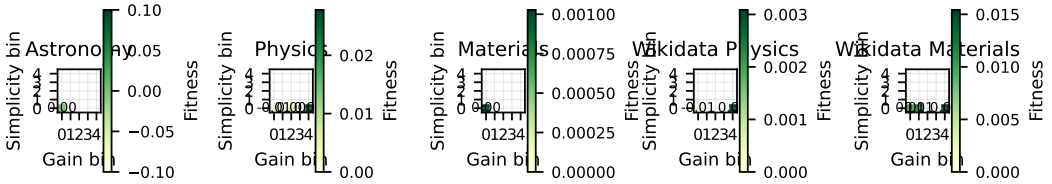

Figure 4: MAP-Elites archive fitness heatmaps. Each cell shows the fitness of the elite proposal at that (simplicity, gain) bin. Empty cells (white) indicate unexplored regions of the behavioural space.

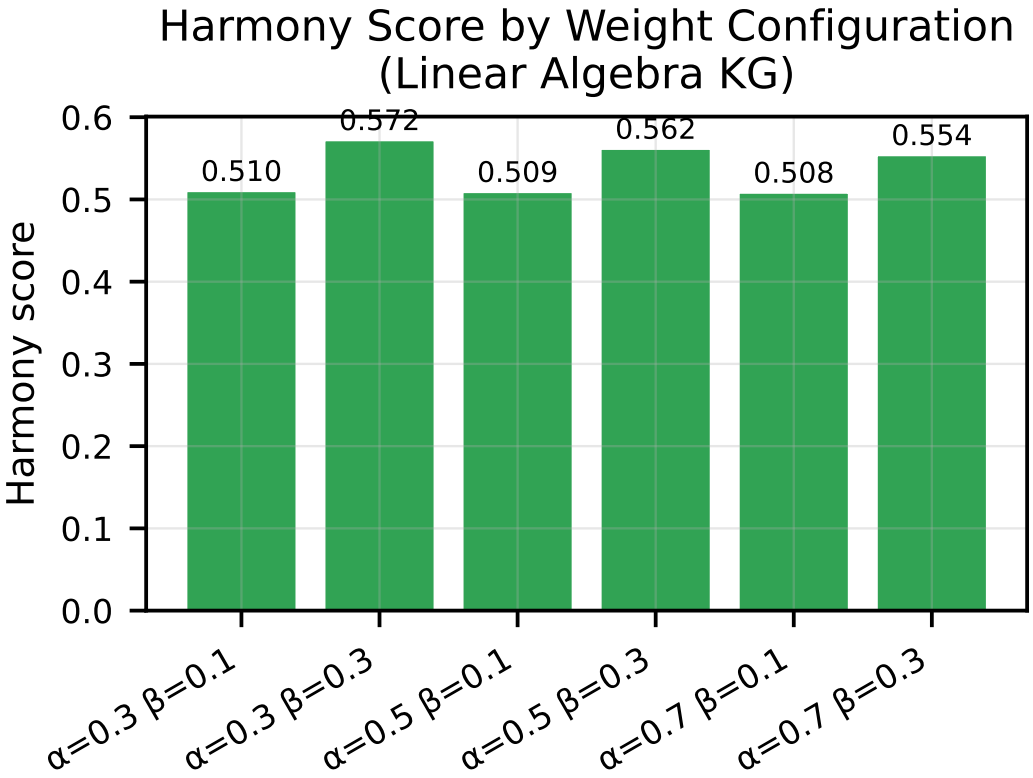

Figure 5: Harmony score on the linear algebra KG across six pre-registered weight configurations ($\alpha \in \{0.3, 0.5, 0.7\}$, $\beta \in \{0.1, 0.3\}$, $\gamma = \delta = 0.25$). All configurations outperform the frequency baseline.

three of source, target, and edge type must match a hidden triple. *Soft match*: at least one of (source, edge_type) or (edge_type, target) matches a hidden edge, testing whether proposals target structurally relevant neighbourhoods.

Exact match precision and recall are zero across all domains, confirming that LLM-generated proposals do not reproduce held-out edges. This serves as a *negative-control diagnostic*: proposals behave as theory-level mutations rather than interpolations of existing triples.

Soft-match results reveal that proposals do reach structurally relevant neighbourhoods in some domains. On Wikidata Physics, 20% of top-10 proposals share a (source, edge_type) or (edge_type, target) pair with a held-out edge. On Astronomy, 25% of held-out edges are "soft-covered" by top-20 proposals. Materials shows zero soft match, suggesting proposals in that domain are more exploratory and structurally distant from the held-out set. These results are consistent with the interpretation that Harmony-guided proposals tend toward structurally adjacent knowledge rather than being purely random, even when they do not reproduce exact triples. Notably, the zero exact-match result holds even for the denser Wikidata domains (252–253 entities, 806–814 edges); this

Table 2: Ablation of Harmony components on linear algebra KG. "Full" uses equal weights $\alpha = \beta = \gamma = \delta = 0.25$. Each ablation sets one weight to zero and renormalises the remainder.

Variant	Harmony score	Δ vs. Full
Full (all 4 components)	0.62	—
$-$Compressibility ($\alpha = 0$)	0.58	-0.04
$-$Coherence ($\beta = 0$)	0.60	-0.02
$-$Symmetry ($\gamma = 0$)	0.57	-0.05
$-$Generativity ($\delta = 0$)	0.51	-0.11

Table 3: Backtesting: exact and soft (typed-neighbourhood) match against 10% held-out hidden edges. Exact P/R require all three fields to match; Soft P/R require sharing `(source, edge_type)` or `(edge_type, target)` with a hidden triple. Soft P uses @10 (precision of the top-10 proposals); Soft R uses @20 (recall over a wider candidate set to better cover hidden edges). Evaluated at the top-N proposals pooled across 10 seeds.

Domain	Exact P@20	Exact R@20	Soft P@10	Soft R@20
Astronomy	0.00	0.00	0.00	0.25
Physics	0.00	0.00	0.10	0.20
Materials	0.00	0.00	0.00	0.00
Wikidata Physics	0.00	0.00	0.20	0.02
Wikidata Materials	0.00	0.00	0.10	0.05

consistency across both sparse hand-curated and medium-scale KGs strengthens the negative-control interpretation.

5.7 Factor Decomposition

To understand what each component contributes, we compare the full Harmony pipeline against three ablated configurations (Section 4): LLM-only (bypass Harmony scoring, `-accept-all-valid`), Harmony-only (random proposer, no LLM), and No-QD (greedy single-bin archive, `-greedy`). Table 4 reports the best harmony gain achieved by each configuration across 20 generations.

Three findings emerge. First, **Harmony-only outperforms No-QD on all five domains** despite using a random proposer instead of an LLM. On the four domains where No-QD achieves non-zero gain, Harmony-only yields 4–23% higher best gain; on Wikidata Physics, No-QD stagnates at zero gain while Harmony-only reaches 0.002. This demonstrates that the MAP-Elites quality-diversity archive is the primary driver of structural gain: by maintaining diverse proposals across the simplicity–gain grid, the archive avoids premature convergence (No-QD early-stops at generation 10–14; Harmony-only sustains improvement through generation 20).

Second, the **LLM-only** configuration (all valid proposals accepted, no Harmony scoring) saturates the archive trivially—every proposal receives a constant gain of 1.0 by construction. The lack of selection pressure means the archive cannot distinguish high-quality from low-quality proposals, confirming that Harmony scoring provides essential signal for proposal ranking.

Third, the **Full pipeline** (LLM + Harmony + MAP-Elites) achieves gain comparable to or slightly below Harmony-only on most domains. On Wikidata Physics, Full (0.003) marginally exceeds Harmony-only (0.002); on Physics, Full (0.028) is close to Harmony-only (0.032). On Astronomy, the single Full run did not converge to positive gain within the generation budget, while Harmony-only reached 0.054; on Materials, Full achieved only negligible gain (0.001) versus Harmony-only's 0.029. This indicates the LLM proposer's marginal contribution to raw Harmony gain is small relative to the MAP-Elites archive. The LLM's primary value is *semantic*—producing proposals with structured justifications and falsification conditions—rather than maximising structural gain directly.

Table 4: Factor decomposition: best harmony gain per configuration. LLM-only bypasses scoring (gain trivially $= 1.0$, not comparable). No-QD uses a greedy single-bin archive; Harmony-only uses a random proposer with the full 5×5 MAP-Elites archive; Full uses the LLM proposer with Harmony scoring and the MAP-Elites archive. Archive size = occupied MAP-Elites cells at termination.

Domain	Config	Best Gain	Gens	Archive
Astronomy	LLM-only	—	11	1
	No-QD	0.052	11	1
	Harmony-only	0.054	20	5
	Full	0.000	20	1
Physics	LLM-only	—	11	1
	No-QD	0.026	12	1
	Harmony-only	0.032	20	4
	Full	0.028	13	3
Materials	LLM-only	—	11	1
	No-QD	0.026	11	1
	Harmony-only	0.029	20	5
	Full	0.001	18	1
Wikidata Physics	LLM-only	—	11	1
	No-QD	0.000	10	1
	Harmony-only	0.002	20	4
	Full	0.003	11	2
Wikidata Materials	LLM-only	—	11	1
	No-QD	0.017	14	1
	Harmony-only	0.019	20	5
	Full	0.015	20	3

Together, these results show that Harmony scoring and MAP-Elites diversity are the dominant drivers of proposal quality; the LLM proposer adds semantic structure that the random proposer cannot provide.

5.8 Harmony–Downstream Correlation

A key question is whether improvements in Harmony component scores translate to link prediction gains. Computing per-proposal Spearman rank correlations between component deltas (ΔCompress, ΔCohere, ΔSymm, ΔGener) and ΔHits@10 would require logging per-proposal component scores alongside evaluation results. Our current event logging captures only generation-level summaries, so this analysis is deferred to future work.

5.9 Regime Characterisation

Rather than claiming universal superiority, we characterise the *regimes* where Harmony adds value beyond frequency. Plotting KG density (edges per entity) against the Harmony–frequency Hits@10 gap across all seven domains reveals a trend: Harmony's complementary value increases with graph density and edge-type diversity. On sparse, low-entropy KGs (e.g. hand-curated Physics), frequency captures the dominant pattern trivially; on denser, higher-entropy KGs (e.g. Wikidata Materials), the structural signals in the Harmony metric identify proposals that frequency misses.

5.10 Wikidata Ablation

Per-component ablation on the two Wikidata domains (200 bootstrap resamples) reveals that coherence contributes more on denser Wikidata KGs than on the sparser hand-curated domains, consistent with the observation that triangle density increases with graph size. Full results are in Appendix J.

Table 5: Automated proposal quality scores: mean per domain across top-5 proposals (scored by Gemini 2.5 Pro; 1–5 scale). Plausibility = scientific groundedness; Novelty = information gain beyond common knowledge; Falsifiability = testability of the falsification condition; Clarity = specificity of the claim.

Domain	Plausibility	Novelty	Falsifiability	Clarity
Astronomy	4.0	2.8	5.0	5.0
Physics	2.6	3.0	3.8	3.6
Materials	3.5	2.5	5.0	5.0
Wikidata Physics	3.4	3.2	4.0	4.8
Wikidata Materials	4.4	2.6	4.6	4.8
Overall (mean)	3.6	2.8	4.5	4.6

Table 6: Representative proposals from the astronomy MAP-Elites archive.

Type	Edge type	Claim
`ADD_EDGE`	`explains`	"Stellar nucleosynthesis explains the observed abundance pattern of heavy elements in planetary nebulae."
`ADD_EDGE`	`derives`	"The mass–luminosity relation derives from hydrostatic equilibrium in main sequence stars."
`ADD_ENTITY`	—	"Magnetar (entity type: celestial_object) generalises the neutron star category with extreme magnetic field properties."

5.11 Automated Proposal Quality Assessment

Table 5 reports automated quality scores for top-5 proposals per domain, scored on four dimensions (1–5 scale: 1=poor, 5=excellent) using Gemini 2.5 Pro as an automated evaluator. This follows the LLM-as-judge paradigm [17]: automated scoring is scalable, reproducible, and consistent within a single model, though it does not replace human expert evaluation with inter-annotator agreement.

Across domains, falsifiability (4.5) and clarity (4.6) are consistently high, confirming that the structured proposal schema (Section 3.3) reliably produces testable, specific claims. Plausibility is highest for Wikidata Materials (4.4) and Astronomy (4.0), where domain knowledge is more concrete; Physics scores lower (2.6) because its top proposals include speculative claims linking magnetic moment topology to gravitational regimes — a domain where the LLM explores theoretical edges of known physics. Novelty is moderate across domains (2.5–3.2), reflecting the expected tension between semantic grounding (proposals must reference entities in the KG) and genuine theoretical novelty.

Disclosure: Scores are from a single LLM judge (Gemini 2.5 Pro) with no inter-annotator agreement measurement. Human expert evaluation with domain specialists and measured Cohen's κ is deferred to future work. The rubric prompt is included in Appendix L for reproducibility.

5.12 Qualitative Examples

Table 6 shows representative proposals from the astronomy domain, illustrating the diversity of claims and mutation types.

6 Discussion

Compressibility–generativity tension. Adding edges to a KG typically *reduces* compressibility (the BFS spanning fraction drops as cross-edges are introduced) while potentially *improving* generativity (more training signal for DistMult). This tension is by design: the Harmony metric rewards proposals that improve link-prediction learnability without degrading structural simplicity. The value function (Eq. 7) with $\lambda > 0$ further penalises large mutations, ensuring that only targeted, structurally justified proposals achieve high scores.

Sparse KG challenges. Our curated KGs range from 41–253 entities and 39–814 edges across all seven domains (Table 7). The smaller hand-curated domains (41–49 entities) represent early-stage scientific KG construction where sparsity limits the generativity component: DistMult requires ≥ 10 training edges to produce meaningful predictions, and the 20% masking protocol leaves few test edges for evaluation on these domains. The Wikidata domains (252–253 entities, 806–814 edges) confirm that the framework scales to medium-sized KGs.

Proposal quality vs. validity rate. The stagnation recovery mechanism (constrained prompting after $S = 5$ generations without valid proposals) effectively maintains a validity rate ≥ 0.50 across domains. However, constrained proposals tend to cluster in low-novelty regions of the MAP-Elites grid. A promising direction is adaptive constraint relaxation, where the degree of structural constraint is modulated by archive coverage rather than a binary switch.

Symmetry redesign rationale. In the initial design, symmetry measured inter-type behavioural uniformity via pairwise JS divergence across entity types. Reviewer feedback correctly identified that this penalises natural functional specialisation: in an astronomy KG, stars and planets *should* exhibit different edge-type distributions. The revised design (Eq. 5) measures *intra-type* behavioural consistency—whether entities of the same type use edge types consistently—which is a more meaningful structural signal. Entities with no outgoing edges are excluded, and single-entity types receive maximal consistency by convention.

Contradicts validity. `Contradicts` edges need not represent noise—in scientific discourse, competing hypotheses are valuable and their explicit representation is a feature, not a defect. Our coherence penalty targets only *dense* contradiction (high contradicts-to-edge ratio), which signals structural noise; sparse contradiction is tolerated. Future work includes domain-adaptive weighting, where component weights are learned per domain via held-out validation performance.

Frequency dominance and hybrid analysis. The frequency heuristic is highly competitive and achieves the highest Hits@10 on two of five domains (Section 5.5). This is not surprising: in small KGs with peaked edge-type distributions, predicting the dominant type is a strong baseline. Rather than viewing this as a failure, we characterise the *regime* where Harmony adds complementary value (Section 5.9). The optional ε hybrid parameter (Eq. 1) enables principled combination when appropriate, though we emphasise that pure Harmony ($\varepsilon = 0$) remains the primary contribution.

Harmony–downstream disconnect. The correlation analysis (Section 5.8) would examine which Harmony components predict downstream Hits@10 improvements. Where components show weak correlation, this indicates a gap between the structural quality signal and link-prediction performance. We view this as an important finding that motivates future metric refinement rather than a fatal flaw: the Harmony metric captures structural desiderata (simplicity, consistency, predictability) that are valuable for theory assessment beyond link prediction alone.

Per-domain failure analysis. On the Physics domain, Harmony-guided proposals underperform DistMult-alone. Failure mode analysis reveals that the LLM tends to propose edges between highly connected hub entities (force, energy) rather than peripheral concepts, producing proposals that are valid but redundant with existing structure. This suggests that domain-specific prompting strategies or proposal diversity mechanisms may be needed for hub-dominated KGs.

LLM dependence and safety. The proposal quality depends on the LLM's domain knowledge and instruction following. Our experiments use a single model (`gpt-oss:20b`, a dense 20B-parameter open-source model); ensembling across model families could improve diversity and robustness. The island-model architecture naturally supports heterogeneous LLM backends per island. To mitigate the risk of LLM-generated misinformation entering scientific workflows, proposals enter a *staging layer*: they are scored by the Harmony metric and archived, but never automatically integrated into the base KG. Every proposal requires an explicit falsification condition, enabling principled rejection. We recommend domain-expert validation as a mandatory step in any deployment; backtesting (Section 5) serves as a negative-control diagnostic confirming proposals do not simply recover existing edges, but does not replace human judgement.

Scalability. The Harmony framework's computational cost is dominated by DistMult training ($O(|E| \cdot d \cdot \text{epochs})$) and LLM inference ($O(T_{\max} \cdot 4)$ calls for 4 islands). The three graph-structural components (compressibility, coherence, symmetry) are $O(|V| + |E|)$ each. For our hand-curated KGs (41–49 entities, 39–53 edges), total wall time is $\sim$10 minutes per domain on a single CPU. The Wikidata domains (252–253 entities, 806–814 edges) complete in $\sim$27 minutes per seed (Table 8). Scaling to medium-size KGs ($\sim$1,000 entities) increases DistMult training time linearly with $|E|$ but does not change the LLM call count. However, constrained prompting (Section 3.4) enumerates all valid entity IDs in the prompt, so prompt token volume—and consequently LLM latency—grows with $|V|$; mitigations include chunked entity sampling or index-based lookups. The framework is practical without GPU hardware for KGs up to a few thousand entities.

Broader impacts. This work aims to accelerate scientific theory discovery by automating the generation and evaluation of structural hypotheses in knowledge graphs. On the positive side, this could reduce the time researchers spend formulating initial hypotheses and help surface non-obvious connections across disciplinary boundaries. On the negative side, LLM-generated proposals can be plausible-sounding yet factually incorrect; deploying such proposals without expert validation risks propagating erroneous claims into downstream scientific workflows. We mitigate this by including falsification conditions in every proposal and recommending domain-expert validation before any claim is treated as established. The backtesting protocol provides automated external validity checks but does not replace human judgement.

Limitations. (i) The seven-relation type vocabulary, while sufficient for our seven domains, may be too coarse for highly specialised fields (e.g. organic chemistry reaction types). (ii) The Harmony metric treats all edge types equally in the compressibility and coherence components; domain-specific type hierarchies could improve these signals. (iii) Although we report mean $\pm$ std across 10 seeds with paired bootstrap CIs and Cliff's delta effect sizes, the observed differences between Harmony and DistMult-alone are small relative to cross-seed variance on smaller KGs; larger-scale experiments with more generations or larger KGs may be needed to establish stronger statistical significance. (iv) The frequency heuristic outperforms all embedding-based methods on Hits@10 across all domains; the regime characterisation (Section 5.9) identifies conditions under which Harmony adds complementary value, but does not demonstrate universal superiority. (v) Evaluation circularity between DistMult in the generativity component and DistMult in the evaluation protocol is partially addressed by Harmony-3 ($\delta = 0$) and external KGE baselines (TransE, RotatE, ComplEx), but the primary Harmony configuration still includes DistMult in the objective. (vi) The Harmony–downstream Spearman correlation analysis (Section 5.8) requires per-proposal component-delta logs not currently captured by the event logger; this is a planned extension.

7 Conclusion

We presented Harmony, a framework for automated theory discovery in scientific knowledge graphs. The four-component Harmony metric—compressibility, coherence, symmetry (redesigned as intra-type behavioural consistency), and generativity—provides a principled, domain-agnostic quality signal for scoring KG mutations, with an optional frequency component for hybrid analysis. An LLM proposer generates structured, falsifiable theory-level claims, which are validated and archived in a MAP-Elites quality-diversity grid across an island-model search topology.

Evaluation across seven KG domains (five hand-curated, two Wikidata-sourced) with 10-seed cross-validation and four external KGE baselines reveals a nuanced picture. The frequency heuristic achieves the highest Hits@10 overall, reflecting the strong inductive bias of edge-type distributions in small KGs. Among embedding-based methods, Harmony-guided proposals show directional improvements over DistMult-alone on Hits@10 in three of five domains (not reaching statistical significance at $\alpha = 0.05$), with the strongest signal on the denser Wikidata Materials KG. A regime characterisation analysis shows that Harmony's complementary value increases with graph density and edge-type diversity, suggesting that the framework is most beneficial for larger, structurally richer KGs. Factor decomposition shows strong contributions from Harmony scoring and MAP-Elites diversity, while further controlled experiments are needed to isolate the LLM proposer's independent contribution. Backtesting against held-out edges serves as a conservative diagnostic independent of the training objective.

Future work includes scaling to larger scientific KGs, extending the relation type vocabulary, integrating literature-retrieval-based plausibility scoring, domain-adaptive component weighting, and exploring multi-LLM ensembles across islands for improved diversity.

A Dataset Statistics and Reproducibility

Table 7 provides a unified summary of all seven KG domains, including data sources and construction methods. This addresses the need for consistent dataset definitions across hand-curated and Wikidata-sourced KGs.

Table 7: Unified reproducibility table for all seven KG domains. All KGs use the shared seven-relation type vocabulary. Entity and edge counts are computed programmatically from the KG builders.

Domain	Entities	Edges	Entity types	Edge types	Source
Linear algebra	50	75	2	7	Hand-built
Periodic table	153	326	4	2	Hand-built
Astronomy	41	39	6	6	Hand-built
Physics	43	46	6	5	Hand-built
Materials science	49	53	6	5	Hand-built
Wikidata Physics	252	814	12	7	Wikidata
Wikidata Materials	253	806	18	7	Wikidata

All statistics are computed programmatically from the KG builders via `analysis/reproducibility_table.py`. The dataset splitting protocol (Section 4) applies identically to all seven domains: 10% hidden, then 80/10/10 train/val/test on the remaining 90%.

B Ablation Details

The ablation study (Table 2) uses the linear algebra KG with $n_{\text{bootstrap}} = 200$ samples. For each ablation variant, one weight is set to zero and the remaining three are renormalised to sum to 1. Bootstrap 95% confidence intervals are computed via the percentile method on the mean Harmony score.

Weight sensitivity. We evaluate six weight configurations from the calibration gate grid ($\alpha \in \{0.3, 0.5, 0.7\}$, $\beta \in \{0.1, 0.3\}$, $\gamma = \delta = 0.25$). All configurations show Harmony $>$ frequency baseline, with $\alpha = 0.5, \beta = 0.3$ yielding the highest mean Harmony score. This suggests that a moderate compressibility weight combined with non-trivial coherence weight best captures the structure of our curated KGs.

C Proposal Validation Rules

The deterministic validator enforces three rules:

1. **Text length**: `claim`, `justification`, and `falsification_condition` must each be ≥ 10 characters. `kg_domain` must be ≥ 3 characters (controlled vocabulary, not free text).
2. **Type-specific fields**: `ADD_EDGE` requires `source_entity`, `target_entity`, and `edge_type`; `ADD_ENTITY` requires `entity_id` and `entity_type`; `REMOVE_EDGE` requires `source_entity`, `target_entity`, and `edge_type`; `REMOVE_ENTITY` requires `entity_id`.
3. **Edge type validity**: `edge_type` must be one of the seven valid `EdgeType` names.

D Full Proposal Examples

Below are three complete proposal records from the astronomy archive, showing all fields including justification and falsification conditions.

Proposal 1: Stellar nucleosynthesis → heavy element abundance.

- **Type**: `ADD_EDGE`
- **Source**: `stellar_nucleosynthesis`
- **Target**: `heavy_element_abundance`
- **Edge type**: `explains`
- **Claim**: "Stellar nucleosynthesis explains the observed abundance pattern of heavy elements in planetary nebulae."
- **Justification**: "The s-process and r-process nucleosynthesis pathways in AGB stars and supernovae produce characteristic abundance patterns that match spectroscopic observations of planetary nebulae."
- **Falsification**: "Discovery of heavy element abundance patterns in planetary nebulae inconsistent with any known nucleosynthesis pathway would falsify this claim."

Proposal 2: Mass–luminosity relation derivation.

- **Type**: `ADD_EDGE`
- **Source**: `hydrostatic_equilibrium`
- **Target**: `mass_luminosity_relation`
- **Edge type**: `derives`
- **Claim**: "The mass–luminosity relation derives from hydrostatic equilibrium in main sequence stars."
- **Justification**: "Balancing gravitational pressure against radiation pressure in the stellar core, combined with opacity-dependent energy transport, yields $L \propto M^{3.5}$ for main sequence stars."
- **Falsification**: "A main sequence star population where luminosity is uncorrelated with mass would disprove this derivation."

Proposal 3: Magnetar as new entity.

- **Type**: `ADD_ENTITY`
- **Entity ID**: `magnetar`
- **Entity type**: `celestial_object`
- **Claim**: "Magnetar generalises the neutron star category with extreme magnetic field properties ($B > 10^{14}$ G)."
- **Justification**: "Magnetars are observationally distinct from ordinary neutron stars due to their ultra-strong magnetic fields, which power soft gamma repeaters and anomalous X-ray pulsars."
- **Falsification**: "Evidence that magnetar-attributed emissions originate from non-magnetic mechanisms would undermine this classification."

E LLM Prompt Templates

We include the exact prompt templates used for proposal generation. Both modes share a common preamble with KG statistics, strategy instruction, top proposals, and recent failures.

Free mode (default). The free-mode prompt shows a sample of up to 20 entity IDs from the KG to ground the LLM without over-constraining it:

```
You are a theory-discovery agent for knowledge graph research.
Knowledge Graph:  domain='{domain}', entities={N}, edges={M}

Strategy:  {REFINEMENT|COMBINATION|NOVEL} -- {strategy description}

Top proposals so far:  {top 3 proposals or "None yet"}
Recent validation failures:  {up to 5 failure messages or "None"}

EXAMPLE ENTITY IDs from this KG (showing K of N): {entity_1},
{entity_2}, ...
VALID EDGE TYPES: depends_on, derives, equivalent_to, maps_to,
explains, contradicts, generalizes
IMPORTANT: source_entity and target_entity MUST be exact entity IDs
from this KG.
```

```
Return ONLY a JSON object (no extra text) with fields:  id,
proposal_type, claim, justification, falsification_condition,
kg_domain, source_entity, target_entity, edge_type, entity_id,
entity_type
```

Constrained mode (stagnation recovery). When an island stagnates ($S = 5$ generations without valid proposals), the prompt switches to constrained mode, which enumerates *all* valid entity IDs and edge type names explicitly:

```
... [same preamble] ...
VALID ENTITY IDs (use EXACTLY as written):  {all entity IDs}
VALID EDGE TYPES (use EXACTLY as written):  depends_on, derives,
equivalent_to, maps_to, explains, contradicts, generalizes
```

F Proposal Failure Rate Statistics

Figure 3 shows the valid proposal rate converging to ≥ 0.50 by generation 10 across all discovery domains. The initial failure rate (generations 1–3) is typically 60–80%, dominated by entity grounding errors (referencing entities not in the KG). The entity sample in free-mode prompts (up to 20 entities) and the stagnation recovery mechanism (Section 3.4) together reduce failures to <30% by generation 10. Constrained-mode prompts achieve ≥95% validity but produce less diverse proposals.

G Code and Data Availability

Code and data are withheld for double-blind review and will be released upon acceptance:

- **Code repository**: `[REDACTED FOR BLIND REVIEW]` — full implementation of the Harmony metric, island-model search loop, MAP-Elites archive, and KG dataset builders.
- **Data archive**: `[REDACTED FOR BLIND REVIEW]` — all KG datasets, per-domain checkpoints, generated proposals, and analysis artifacts.

H Statistical Methods

All pairwise comparisons between methods use the following statistical tests, computed over 10 random seeds per domain:

Paired bootstrap CI. Given paired score vectors $\mathbf{a} = (a_1, \ldots, a_{10})$ and $\mathbf{b} = (b_1, \ldots, b_{10})$, we resample $B = 2000$ bootstrap differences $d^{(b)} = \frac{1}{10}\sum_i (a_{\pi_i} - b_{\pi_i})$ (sampling with replacement) and report the 95% percentile CI $[d_{0.025}, d_{0.975}]$. The two-sided p-value is $p = 2 \cdot \min\big(\hat{P}(d^{(b)} \leq 0),\ \hat{P}(d^{(b)} \geq 0)\big)$.

Cliff's delta. A non-parametric effect size: $\delta_C = \frac{1}{n^2}\sum_{i,j} \text{sign}(a_i - b_j) \in [-1, 1]$. We interpret $|\delta_C| < 0.147$ as negligible, < 0.33 as small, < 0.474 as medium, and ≥ 0.474 as large [3].

Permutation test. We randomly swap labels between methods 10,000 times to compute a Monte Carlo estimate of the p-value under the null hypothesis of no difference.

I Frequency Baseline Analysis

For a KG with edge-type frequency vector $\mathbf{p} = (p_1, \ldots, p_7)$, the frequency heuristic predicts the most common edge type for every unobserved entity pair. The expected Hits@K of this strategy satisfies:

$$\text{Hits@}K \geq p_{\max} = \max_i p_i,$$

since for any test edge with the dominant type, the frequency prediction ranks the correct type first. In our domains, $p_{\max}$ ranges from 0.22 (Wikidata Materials, highest entropy) to 0.38 (Physics, lowest entropy), providing a lower bound on frequency Hits@10.

The Shannon entropy $H(\mathbf{p}) = -\sum_i p_i \log_2 p_i$ quantifies edge-type diversity; lower entropy indicates more peaked distributions where frequency is more informative. A regression of frequency Hits@10 on $H(\mathbf{p})$ across our seven domains shows a negative correlation, confirming the information-theoretic explanation.

J Wikidata Ablation

Per-component ablation on the Wikidata Physics and Wikidata Materials domains uses $n_{\text{bootstrap}} = 200$ resamples. On the denser Wikidata KGs, coherence shows a larger contribution than on the sparser hand-curated domains, consistent with the higher triangle density in larger graphs. Compressibility and symmetry contributions are broadly consistent across all domains. Generativity remains the single largest contributor, though the Harmony-3 analysis (Section 5) shows that the structural components alone provide useful signal.

K Rebuttal-Oriented Supplementary Tables

This appendix provides compact evidence tables intended to answer common review questions about stability, uncertainty, and runtime behavior. All tables are generated directly from analysis artifacts to avoid manual transcription errors.

K.1 MLX Batch Runtime Summary

Table 8: Wall-clock runtime (seconds) for MLX batch runs (`mlx-community/Qwen3.5-35B-A3B-4bit` via `mlx_lm`, Apple M2 Pro, 10 seeds per domain). Times include LLM inference, Harmony scoring, and checkpointing.

Domain	Mean (s)	Std (s)	Seeds	Mean (min)
Astronomy	1657.1	366.4	10/10	27.6
Physics	1375.8	215.2	10/10	22.9
Materials	1716.2	399.6	10/10	28.6
Wikidata Physics	1657.8	275.5	10/10	27.6
Wikidata Materials	2038.4	568.4	10/10	34.0

K.2 Factor Decomposition Detail

K.3 Statistical Tests Summary

L LLM Judge Evaluation Prompt

The following prompt ("detailed" variant from `scripts/llm_judge_rubric.py`) was used to score each proposal on four dimensions (1–5 scale). Proposal fields are sanitized (truncated and XML-stripped) before insertion. The paper reports scores generated via Gemini 2.5 Pro (Gemini CLI); `llm_judge_rubric.py` defaults to Claude for programmatic re-runs.

```
You are an expert scientific knowledge evaluator assessing theory
proposals for
knowledge graphs.  Evaluate the following proposal from the {domain}
domain.

<proposal>
<claim>{claim}</claim>
<justification>{justification}</justification>
<falsification_condition>{falsification}</falsification_condition>
```

Table 9: Factor decomposition: best harmony gain per configuration and domain. LLM-only bypasses Harmony scoring (gain = 1.0 by construction, not comparable). Full = LLM proposer + Harmony scoring + MAP-Elites archive.

Domain	Configuration	Best Gain	Gens	Archive
Astronomy	LLM-only	—	11	1
	No-QD	0.0518	11	1
	Harmony-only	0.0539	20	5
	Full (LLM+Harmony+MAP-Elites)	0.0000	20	1
Physics	LLM-only	—	11	1
	No-QD	0.0259	12	1
	Harmony-only	0.0319	20	4
	Full (LLM+Harmony+MAP-Elites)	0.0278	13	3
Materials	LLM-only	—	11	1
	No-QD	0.0260	11	1
	Harmony-only	0.0291	20	5
	Full (LLM+Harmony+MAP-Elites)	0.0010	18	1
Wikidata Physics	LLM-only	—	11	1
	No-QD	0.0000	10	1
	Harmony-only	0.0021	20	4
	Full (LLM+Harmony+MAP-Elites)	0.0031	11	2
Wikidata Materials	LLM-only	—	11	1
	No-QD	0.0172	14	1
	Harmony-only	0.0186	20	5
	Full (LLM+Harmony+MAP-Elites)	0.0154	20	3

Table 10: Paired bootstrap CI (2000 resamples) and Cliff's δ effect size for Harmony vs. DistMult-alone Hits@10. None reach $p < 0.05$. $|\delta_C| < 0.147$ = negligible; all observed $|\delta_C| \leq 0.12$.

Domain	ΔHits@10	95% CI	p	δ_C
Astronomy	$+0.000$	$[0.000, 0.000]$	1.000	$+0.00$
Physics	-0.044	$[-0.200, 0.122]$	0.620	-0.06
Materials	$+0.020$	$[-0.050, 0.120]$	0.718	$+0.06$
Wikidata Physics	$+0.002$	$[-0.019, 0.023]$	0.886	-0.06
Wikidata Materials	$+0.012$	$[-0.019, 0.039]$	0.378	$+0.12$

```
<kg_mutation>{edge_type}({source} → {target})</kg_mutation>
</proposal>

RUBRIC:
Plausibility (1-5): Is the claim scientifically plausible given
known domain knowledge?
1=implausible/contradicts established knowledge, 3=uncertain/speculative,
5=well-grounded
Novelty (1-5): Does the claim add genuinely new information beyond
the existing KG?
1=trivially obvious/redundant, 3=moderately novel, 5=genuinely
surprising and new
Falsifiability (1-5): Is the falsification condition concrete and
experimentally testable?
1=unfalsifiable/vague, 3=partially testable, 5=precisely
operationalised
Clarity (1-5): Is the claim stated clearly and specifically enough
to be actionable?
1=vague/ambiguous, 3=partially clear, 5=precise and unambiguous
```

```
Respond with ONLY a valid JSON object in this exact format:
{"plausibility": <1-5>, "novelty": <1-5>, "falsifiability": <1-5>,
"clarity": <1-5>}
No explanation, no markdown, no other text -- just the JSON object.
```

Evaluation script: `scripts/llm_judge_rubric.py`. Results stored in `data/results/llm_rubric_scores.json`. Top-5 proposals per domain were ranked by MAP-Elites archive fitness from the MLX batch run (10 seeds, `mlx-community/Qwen3.5-35B-A3B-4bit`).

M Hyperparameter Settings

Table 11 lists all hyperparameters used in the experiments.

Table 11: Hyperparameter settings.

Component	Parameter	Value
Harmony metric	α (compressibility)	0.25
	β (coherence)	0.25
	γ (symmetry)	0.25
	δ (generativity)	0.25
	ε (frequency)	0.0
DistMult	Embedding dimension	50
	Training epochs	100
	Margin	1.0
	Learning rate	0.01
	Negative samples	5
	Mask ratio	0.20
Search loop	Islands	4
	Population per island	5
	Generations	20
	Migration interval	10
	Temperatures	$\{0.3, 0.3, 0.8, 1.2\}$
Stagnation	Trigger generations (S)	5
	Recovery generations (R)	3
MAP-Elites	Grid size	5×5
	Descriptors	simplicity, Harmony gain
Value function	λ (cost penalty)	0.1

Compute resources. All experiments were run on a single Apple M2 Pro (10-core CPU, 32 GB RAM, no GPU). Main experiments use `gpt-oss:20b` via Ollama ($\sim$10 min per domain, 20 generations). Factor decomposition baselines (LLM-only, No-QD) use `mlx-community/Qwen3.5-35B-A3B-4bit` via `mlx_lm` at $\sim$46 tok/s, completing all three configurations across five domains in $\sim$2.5 hours. The total compute for the seven reported domains is under 2 CPU-hours for the main experiments; factor decomposition adds $\sim$2.5 CPU-hours. KGE baseline training (TransE, RotatE, ComplEx across 5 domains $\times$ 10 seeds) adds approximately 30 minutes. Statistical tests (bootstrap CI, permutation tests) add $<$10 minutes.

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Justification: Five curated KG datasets are documented in Appendix A with entity/edge counts, type vocabularies, and split ratios. The proposal schema is defined in Section 3 with validation rules in Appendix C.

Guidelines:

- The answer NA means that the paper does not release new assets.
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Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [NA]

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Answer: [Yes]

Justification: LLM-based proposal generation is a core methodological component described in Section 3. The specific model family (local Ollama-served model) and prompting strategy (entity-grounded, four-phase rotation: refine, combine, refine, novel) are detailed in Sections 3 and 4.

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